Recod.ai/LUC - Scientific Image Forgery Detection

Max Bakke Seymour[1], Einar Martin Søvik Bernsen[1], and Emre Balci[1]

[1]University of Bergen (UiB)

Abstract

We present a pipeline for scientific image copy-move forgery detection using a DINO-based feature extractor and segmentation decoder. In addition, we implement a SegNeXt-inspired segmentation baseline to evaluate whether learned approaches can compete with feature-based pipelines. The DINO model achieves a validation F1 score of 0.1850, while the best SegNeXt-based model reaches 0.1827.

1 Introduction

Copy-move forgery detection aims to identify duplicated regions within an image. This task is particularly challenging in scientific images due to repeated patterns and homogeneous backgrounds.

2 Methods

2.1 Dataset

We use the Recod.ai/LUC dataset with an 80/10/10 split. Images are resized to 448×448.

2.2 DINO Baseline

We use a frozen DINOv2 backbone to extract features and a lightweight decoder to produce pixel-level predictions.

2.3 Post-processing

We apply Gaussian smoothing, thresholding, and morphological operations to refine predicted masks.

2.4 SegNeXt-inspired Model

We implement a convolutional encoder-decoder architecture. A pretrained `convnext_tiny` encoder is used in the best configuration.

3 Results

Model	Pretrained	Train Size	Epochs	Val F
DINO	Yes	900	8	**0.185**
SegNeXt (custom)	No	500	5	0.083
SegNeXt (pretrained)	Yes	900	12	0.182

Table 1. Model comparison

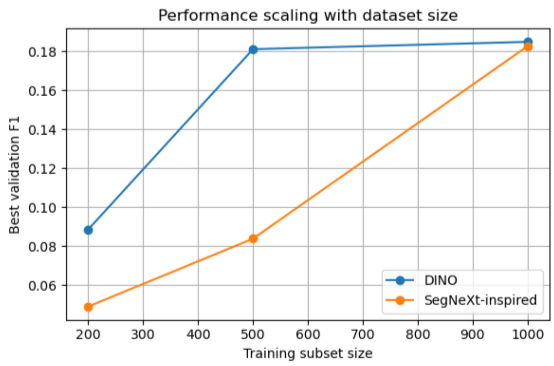

Figure 1. Performance comparison between models

The DINO model achieves the highest performance. The pretrained SegNeXt model performs similarly, while the non-pretrained version performs significantly worse.

4 Discussion

The results show that pretrained representations are critical. Without them, segmentation models fail to learn meaningful features. The DINO pipeline is more stable and more data-efficient.

5 Conclusion

Segmentation-based approaches can be competitive, but strong pretrained features are required. Future work should focus on improving alignment between training objectives and evaluation metrics.